

Name: _____

Period: _____

Wed, Sept 23 (Pd 6) / Thu, Sept 24 (Pd 1, 3)

Warmup

1. An airplane is flying with a horizontal component velocity of 247 m/s and a vertical component velocity of 166 m/s. What is the resultant velocity of the airplane, and at what angle is it flying? *Make sure to draw a triangle.*

2. To throw a runner out at home plate, a player throws a baseball with a velocity of 41 m/s at a 36-degree angle. What are the horizontal and vertical components of his throw? *Make sure to draw a triangle.*

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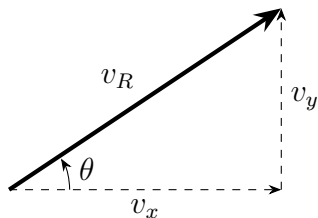
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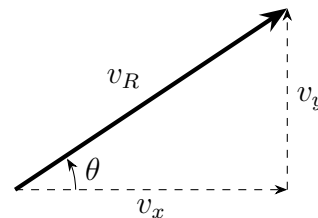
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Equations



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$$v_x^2 + v_y^2 = v_R^2$$
$$v_x = v_R \cos(\theta)$$
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